



Nanoplastics exacerbate Parkinson's disease symptoms in *C. elegans* and human cells

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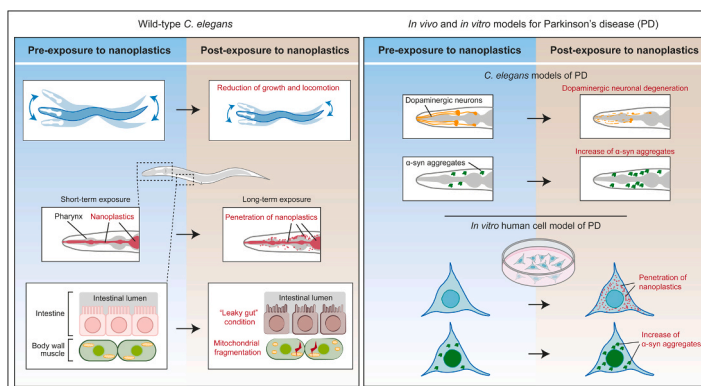
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HIGHLIGHTS

- Low concentrations of nanoplastics inhibit the growth and movement of nematodes.
- Nanoplastics induce leaky gut and deep tissue penetration in *C. elegans*.
- Nanoplastics increase α -Synuclein aggregation.
- The effects vary depending on the surface groups of the nanoplastics.

GRAPHICAL ABSTRACT



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ABSTRACT

The increasing prevalence of nanoplastics in our environment due to the widespread use of plastics poses potential health risks that are not yet fully understood. This study examines the physiological and neurotoxic effects of these minuscule nanoplastic particles on the nematode *Caenorhabditis elegans* as well as on human cells. Here, we find that 25 nm polystyrene nanoplastic particles can inhibit animal growth and movement at very low concentrations, with varying effects on their surface groups. Furthermore, these nanoplastic particles not only accumulate in the digestive tract but also penetrate further into extraintestinal tissues. Such nanoplastics significantly compromise the integrity of the intestinal barrier, leading to "leaky gut" conditions and cause mitochondrial fragmentation in muscles, which possibly explains the observed movement impairments. A striking discovery was that these nanoplastics exacerbate symptoms similar to those of Parkinson's disease (PD), including dopaminergic neuronal degeneration, locomotor dysfunction, and accumulation of α -Synuclein aggregates. Importantly, our study demonstrates that the detrimental effects of nanoplastics on the aggregation of

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α -Synuclein extend to both *C. elegans* and human cell models of PD. In conclusion, our research highlights the potential health hazards linked to the physicochemical properties of nanoplastics, underlining the urgency of understanding their interactions with biological systems.

Environmental implication: The escalating prevalence of nanoplastics in the environment due to widespread plastic usage raises potential health risks. Studies conducted on *C. elegans* indicate that even low concentrations of 25 nm polystyrene nanoplastics can impair growth and movement. These particles accumulate in the digestive system, compromising the intestinal barrier, causing “leaky gut”, as well as inducing Parkinson’s-like symptoms. Importantly, in both *C. elegans* and human cell models of Parkinson’s disease, such nanoplastics penetrate tissues or cells and increase α -Synuclein aggregates. This underscores the urgent need to understand the interactions of nanoplastics with biological systems and highlights potential environmental and health consequences.

1. Introduction

The global rise in plastic production and consumption has led to an alarming increase in environmental contamination [16,39]. When plastic waste is exposed to biological, chemical, and environmental factors, it breaks down due to mechanical and chemical processes, like hydrolysis and UV radiation [26,3,51]. This results in the formation of microplastic (100 nm–5 mm) and nanoplastic (less than 100 nm) particles [15,3,46]. Consequently, micro- and nano-plastics have become increasingly prevalent in various ecosystems, posing potential risks to both wildlife and humans [40,63].

When organisms consume micro- or nano-plastics, these particles can accumulate in their digestive systems and potentially infiltrate other tissues [4]. Notably, nanoplastics are so minuscule that they can easily penetrate tissues. They have been found in blood, breast milk, placentas, infant feces, and even in the brain, where they can breach the blood-brain barrier [24,27,33,45]. Some studies have suggested potential links between nanoplastics and neurological impairments such as memory loss, cognitive dysfunction, and emotional/social behavior [28, 41,50]. These issues are becoming closely associated with neurological disorders, including Parkinson’s disease (PD), Alzheimer’s disease, or autism spectrum disorders [35,49,59,64]. However, the exact neurological conditions caused by nanoplastics remain elusive.

PD is a fast-growing neurodegenerative disorder characterized by the progressive loss of dopaminergic neurons in the substantia nigra of the midbrain, primarily affecting movement [6]. A hallmark of PD is the accumulation of abnormal protein aggregates, particularly α -Synuclein, which form structures called Lewy bodies [17]. These Lewy bodies are pathological inclusions found in the brains of people with PD. The development of PD is influenced by a combination of genetic and environmental factors. Growing evidence suggests that exposure to environmental hazards, such as pesticides or chemicals, contributes to an increased risk or progression of PD [5]. Recently, emerging research suggests that the gut-brain axis, a bidirectional communication system between the digestive system and the brain, may play a role in PD [23]. Tiny plastic particles often accumulate in the intestine, which could compromise the integrity of the intestinal barrier [31,61]. The breach in the intestinal barrier by these particles may permit harmful substances to reach the brain, amplifying neuroinflammation and neuronal damage [18,57,62]. Previous research has suggested that nanoplastics could influence gene expression in mouse neurons associated with PD [35]. However, the effects and mechanisms of these nanoplastic particles on the development of PD are still uncertain.

In this study, we report the physiological and neurotoxic effects of nanoplastic particles based on their physical and chemical properties, including concentration and surface charge. We conduct tests on these effects because of the possibility that weathered plastics could acquire surface charges, potentially influencing bioaccumulation and health. To assess our hypothesis, we examine the effects of nanoplastics on the physiology and behavior of *C. elegans*. Our results indicate that nanoplastic particles, while exhibiting minimal acute systemic or reproductive toxicity, significantly affect the multifaceted physiology of *C. elegans* at even very low concentrations. Such particles not only hinder worm growth and movement but also accumulate in the digestive tract,

penetrate deeper into extraintestinal tissues, lead to “leaky gut” conditions, and cause muscle mitochondrial fragmentation. Moreover, these nanoplastics exacerbate the symptoms associated with Parkinson’s pathology in *C. elegans* models of PD. They also have a similar effect on human cells. We propose that the effects of nanoplastics vary depending on their chemical properties and the specific phenotypes of interest. Our research is among the pioneering efforts to highlight the potential neurotoxic risks associated with the physicochemical properties of nanoplastics.

2. Materials and methods

2.1. *C. elegans* strains

All strains were cultured at a temperature of 20 °C. Worms were grown on NGM agar plates or in complete K-medium [8,53]. The strains used include the wild type (N2) and reporter strains, such as SD1347 (*ccIs4251[myo-3p::mitochondrial GFP; myo-3p::GFP::LacZ::NLS; dpy-20(+)]*), CL2166 (*dIs19[gst-4p::GFP::NLS]*), UA44 (*baIn11[dat-1p:: α -Synuclein; dat-1p::GFP]*), and NL5901 (*pkIs2386[unc-54p:: α -Synuclein::YFP; unc-119(+)]*). Some strains were provided by the CGC, which is funded by NIH Office of Research Infrastructure Programs (P40 OD010440). The UA44 strain was a kind gift from Kim Caldwell’s laboratory at the University of Alabama. All worms were fed *E. coli* OP50 as their food source.

2.2. Nanoplastic particles

Polystyrene nanoplastic (PS-NP) beads with diameters of 25 nm featuring different surface groups, including pristine PS, carboxyl (COOH)-PS, and primary amino (NH₂)-PS were purchased from Lab261 (USA) (catalog numbers PST25, PST25C, and PST25A, respectively). Red fluorescent 25 nm PS beads including pristine PS, carboxyl (COOH)-PS, and primary amino (NH₂)-PS were also purchased from Lab261 (USA) (catalog numbers FRP25, FRP25C, and FRP25A, respectively). All PS-NP beads were diluted in Complete K-medium for *C. elegans* experiments and in 1X PBS (Phosphate Buffered Saline) for cell experiments, then vortexed prior to use.

2.3. Organism-level assays in *C. elegans*

2.3.1. Synchronizing developmental stages of *C. elegans*

The P₀ generation of *C. elegans* is cultivated on NGM plates with OP50 for 4 days. Following this, the F₁ generation embryos are isolated using a bleaching assay [53]. These embryos are then rinsed four times with M9 buffer before being cultured either on NGM agar plates or in complete K-medium, all maintained at 20 °C.

2.3.2. Body length measurement

Worms at the L4 stage were exposed to PS-NP beads in complete K-medium at the concentration of 10, 100, or 1000 μ g/L for a day. After this treatment, the worms were rinsed with complete K-medium and left to dry on NGM plates for 20 min. Individual worms were then gently placed on a 10% agar pad on a glass slide. To prevent them from drying

out, 2.5 μ L of M9 buffer was added. Observations were made using a Leica DM2000 compound microscope, which was equipped with a Leica DFC7000 GT camera. The length of each worm was measured from its head to its tail along its central axis using the ImageJ software. The experiments were repeated 7 times independently for each condition.

2.3.3. Survival assay

Approximately 40 L4 stage worms were subjected to PS-NP beads exposure in complete K-medium at various concentrations for a day. To evaluate lethality, worms were examined for movement after shaking. Any worm that did not move was considered lethal. Experiments were repeated 4 times independently.

2.3.4. Brood size assay

Worms at the L1 stage were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L for 2 days. After the exposure period, the worms were washed with complete K-medium and then placed on NGM plates. Each individual worm was then isolated and transferred daily to fresh NGM plates. The number of offspring produced by each worm was counted when the offspring reached to the late larval stage. Experiments were repeated 3 times independently.

2.4. Cell and tissue-level assays in *C. elegans*

2.4.1. Tissue penetration in *C. elegans*

In the short-term exposure experiment, day 3 adult worms were treated with PS-NP beads in complete K-medium at a concentration of 15 mg/L for 1 day (resulting in day 4 adults). In the long-term exposure experiment, worm embryos were exposed to PS-NP beads in complete K-medium at a concentration of 15 mg/L for 7 days (resulting in day 4 adults). The liquid culture media was refreshed every 2 days starting 4 days after the exposure to prevent crowding by progeny. Subsequently, the worms were washed with complete K-medium and allowed to dry on NGM plates for 20 min. Each worm was then carefully placed on a 10% agar pad on a glass slide, with the addition of 2.5 μ L of 1 μ M levamisole for anesthesia. Images were captured using a Carl Zeiss LSM710 confocal microscope at a resolution of 1024 $\times$ 1024 with a Z-stack slice thickness of 2 μ m at an average of 2. The Zen Black 2011 SP2 software was employed for image acquisition. Prior to analysis, the images were processed in ZEN 3.6 (Blue edition) by applying maximal projection. Experiments were repeated 3 times independently.

2.4.2. Intestinal barrier function assay

Worm embryos were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L for 7 days (resulting in day 4 adults). Subsequently, these worms were washed with complete K-medium and allowed to dry on NGM plates for 20 min. Approximately 10 worms were then incubated at 20 °C for 3 h in a liquid culture containing 30 μ L of OP50 bacteria mixed with Blue No.1 food dye (FD&C) [29]. Following this incubation, each worm was transferred to an unseeded NGM plate. Each individual worm was placed on a 10% agar pad on a glass slide, with the addition of 2.5 μ L of M9 buffer. Observations were conducted a Leica DM2000 compound microscope. Experiments were repeated 4 times independently.

2.4.3. Mitochondrial fragmentation in muscle

Embryos of the SD1347 worm strain were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L for 7 days (resulting in day 4 adults). Subsequently, the worms were washed with complete K-medium and dried on NGM plates for 20 min. Each worm was then placed on a 10% agar pad on a glass slide, with the addition of 2.5 μ L of M9 buffer. The worms were observed by using a Leica DM2000 compound microscope. Experiments were repeated 3 times independently.

2.4.4. GST-4::GFP reporter assay

Embryos of the CL2166 worm strain were exposed to PS-NP beads in

complete K-medium at a concentration of 15 μ g/L for 7 days (resulting in day 4 adults). Subsequently, the worms were washed with complete K-medium and dried on NGM plates for 20 min. 10 worms were then placed on a 10% agar pad on a glass slide, with the addition of 1 μ L of 5 μ M levamisole for anesthesia. The worms were observed by using a Leica DM2000 compound microscope equipped with a Leica DFC7000 GT camera. The intensity of each worm was measured by using ImageJ software. Experiments were repeated 3 times independently.

2.5. Neuronal assays in *C. elegans*

2.5.1. Thrashing assay

For short-term exposure, L4 stage worms were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L for 1 day (resulting in day 1 adults). For long-term exposure, worm embryos were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L, lasting either 5 days (resulting in day 2 adults) or 7 days (resulting in day 4 adults). In the long-term exposure group, the liquid culture medium was replaced every 2 days, starting from the fourth day post-exposure, to prevent overcrowding from offspring.

To assess the worm's thrashing behavior, each worm was placed onto a small droplet of M9 buffer, approximately 2 μ L. A dissecting microscope (Leica MSV269) equipped with a camera (Leica K3M) was used to record the worm's movement for 30 s. These recordings were then analyzed using either manual tracking or the wrMTrck multiple object tracker plug-in in ImageJ (version 2.9.0). During the wrMTrck analysis, the following settings were applied: minsize= 100, maxsize= 2500, maxvelocity= 50, maxareachange= 210, mintracklength= 50, bend-threshold= 1, and binsize= 0. Any thrashing movement, defined by the worm bending either to its left or right, was counted as a single unit of movement. Experiments were repeated 3 times independently.

2.5.2. Observation of dopaminergic neuronal morphology

Embryos from the UA44 worm strain were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L for 7 days (resulting in day 4 adults). Subsequently, the worms were washed with complete K-medium and then allowed to dry on NGM plates for 20 min. Each worm was then carefully placed on a 10% agar pad on a glass slide, with the addition of 2.5 μ L of M9 buffer. The worms were observed by using a Leica DM2000 compound microscope. Experiments were repeated 3 times independently.

2.5.3. α -Synuclein aggregation assay in *C. elegans*

Embryos of the NL5901 worm strain were exposed to PS-NP beads in complete K-medium at a concentration of 15 μ g/L for 7 days (resulting in day 4 adults). The liquid culture media was refreshed every 2 days starting 4 days after the exposure to prevent crowding by progeny. The worms were then imaged with a confocal microscope as described in 2.5.1. Experiments were repeated 4 times independently.

2.6. Human cell assays

2.6.1. Human cell line culture

α -Synuclein (A53T)::EGFP overexpressing SH-SY5Y cells (A53T α -syn-EGFP SH-SY5Y cells) were generously provided by Sang Myun Park from Ajou University School of Medicine. These cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum and maintained at 37 °C in a humidified atmosphere containing 5% CO₂ and 95% air.

2.6.2. Preparation of α -Synuclein pre-formed fibrils (PFFs)

α -Synuclein PFF was prepared as described previously using lyophilized recombinant human monomeric α -Synuclein (rPeptide, S-1001–2) [9]. Briefly, monomeric α -Synuclein was dissolved in sterile PBS to a final concentration of 5 mg/ml, followed by incubation on an orbital shaker (Thermomixer F1.5) at 37 °C with continuous agitation at 2000 x

g for 7 days, and then stored at -80°C until use as α -Synuclein PFF. The status of α -Synuclein PFF was determined using the thioflavin T binding assay. α -Synuclein PFFs were sonicated on ice for 1 min (1 s pulse on/off) at 20% amplitude using an ultrasonic processor VC 505 before use.

2.6.3. Cell penetration of PS-NP and α -Synuclein aggregation assay in human cell model of PD

A53T α -syn-EGFP SH-SY5Y cells were seeded onto coverslips and treated with α -Synuclein PFF in the presence or absence of PS beads for 1 day. For monitoring of cell penetration of PS-NP, RF-PS beads were used. The cells were washed with PBS and fixed in 4% paraformaldehyde for 20 min at room temperature. Subsequently, they were washed three times in 1X PBS and permeabilized with 0.1% Triton-X 100 for 10 min at room temperature. After three washes in 1X PBS, the nuclei were stained with a 300 nM DAPI solution for 10 min. Finally, the cells were washed three times in 1X PBS, rinsed with distilled water, and mounted. Images were captured using a Leica confocal microscope at a resolution of 1024×1024 with a Z-stack slice thickness of $0.4 \mu\text{m}$, averaging 12 slices. Three random fields were selected, and more than 120 cells per group were analyzed in each experiment. MetaMorph software (versions 7.7.8.0 or 7.10.4.407) was used to calculate the

number of α -Synuclein aggregates per cell. Experiments were independently repeated 3 times, and the number of α -Synuclein aggregates was analyzed in a total of 9 fields.

2.7. Statistical analysis

Statistical analyses were conducted using the GraphPad Prism software (version 5.0). The significance of differences was determined using unpaired *t*-tests for two samples, one-way ANOVA followed by Tukey's multiple comparison tests for multiple samples, and Fisher's exact test for two categorical variables. $p < 0.05$ (*) was considered statistically significant. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns = not significant. Data are represented as mean $\pm$ SEM, unless otherwise stated. The experiments shown in this study were conducted independently 3 to 7 times. Details regarding specific statistical analyses, *p* values, statistical significance, and sample sizes for all graphs can be found in the accompanying figures or figure legends. Unless otherwise specified, 'n' represents the number of animals tested, and 'N' denotes the number of biological replicates.

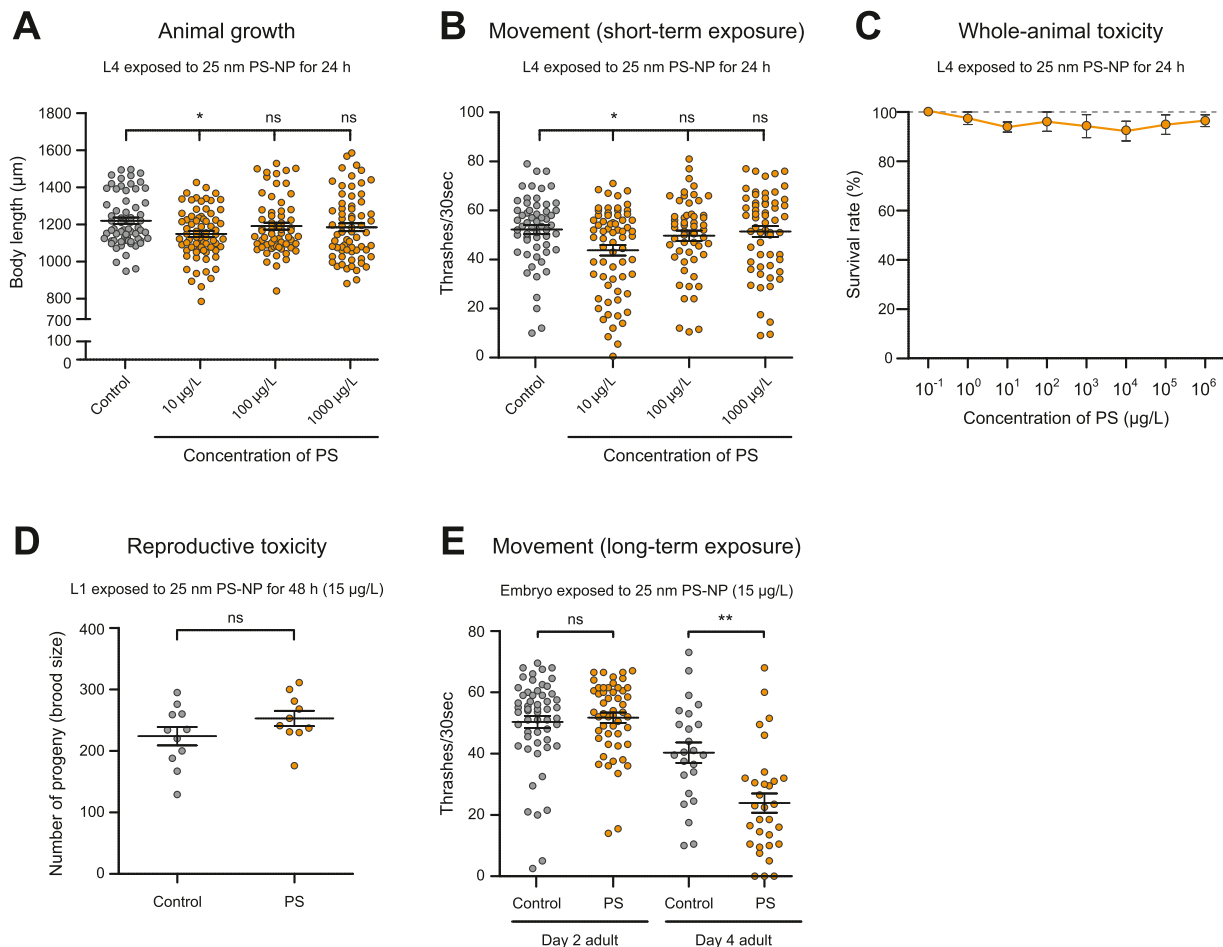


Fig. 1. Effects of exposure to 25 nm polystyrene nanoplastics (PS-NP) on the physiology and behavior of *C. elegans*. (A) Body length of wild type *C. elegans* at the L4 stage after exposure to different concentrations of 25 nm PS-NP beads. Statistics: one-way ANOVA. $N = 7$. $n = 64, 72, 63$, and 66 worms per group, respectively. (B) Thrashing frequency of wild type *C. elegans* at the L4 stage after exposure to different concentrations of PS-NP. Statistics: one-way ANOVA. $N = 7$. $n = 58, 67, 55$, and 59 worms per group, respectively. (C) Survival rate of wild type *C. elegans* at the L4 stage after exposure to different concentrations of PS-NP. Statistics: one-way ANOVA: not significant. $N = 4$. $n = 153, 168, 140, 178, 151, 148, 142$, and 98 worms per group, respectively. (D) Brood size of wild type *C. elegans* at the L4 stage after exposure to PS-NP at a concentration of $15 \mu\text{g/L}$. Statistics: unpaired *t*-test. $N = 3$. $n = 11$, and 10 worms per group, respectively. (E) Thrashing frequency of wild type *C. elegans* embryos exposed to PS-NP for either 5 days (reaching day 2 adulthood) or 7 days (reaching day 4 adulthood). Statistics: one-way ANOVA. $N = 3$. $n = 55, 51, 24$ and 31 worms per group, respectively.

3. Results

3.1. Physiological and locomotive effects of polystyrene nanoparticle exposure in *C. elegans*

In this study, we used round polystyrene nanoplastic (PS-NP) beads with a diameter of 25 nm to assess the physiological effects of nanoplastic exposure. We first investigated the concentration-dependent effects of nanoplastic exposure on *C. elegans*. Intriguingly, the most dramatic reduction was observed at a lower concentration of 10 $\mu\text{g/L}$, as compared to the effects at 100 and 1000 $\mu\text{g/L}$ of PS-NP concentrations (Fig. 1A). Moreover, behavioral assays capturing thrashing ('lateral swimming') movements also showed a lower concentration of 10 $\mu\text{g/L}$ produced the most dramatic reduction of thrashing frequency (43.8 ± 2.1 per 30 s), as compared to the effects at 100 and 1000 $\mu\text{g/L}$ of concentrations (49.7 ± 2.1 and 51.4 ± 2.2 per 30 s, respectively) (Fig. 1B).

We further explored the toxicity of 25 nm PS-NP on *C. elegans*. Despite a 24-hour exposure at the fourth larval stage (L4) to very high concentrations up to 10^6 $\mu\text{g/L}$, there was no significant decrease in survival (Fig. 1C). We then assessed the reproductive toxicity by measuring the brood size after exposing L1 stage worms to 25 nm PS-NP at a concentration of 15 $\mu\text{g/L}$ for 48 h. We observed no significant alterations (Fig. 1D).

We then studied the impact of chronic exposure to 25 nm PS-NP on *C. elegans* locomotion. Worms were exposed to the beads from the embryo stage for durations of 5 and 7 days, after which thrashing frequency was measured. Notably, worms exposed for 7 days (resulting in day 4 adults) exhibited a substantial decrease in thrashing, while those exposed for 5 days (resulting in day 2 adults) did not show a similar reduction (Fig. 1E).

As a result, we showed that 25 nm PS-NP at the low concentrations (i.e., 10 or 15 $\mu\text{g/L}$) resulted in reductions in animal growth and mobility. However, these nanoparticles did not exhibit marked whole-animal or reproductive toxicity. Importantly, prolonged exposure to nanoplastics showed more pronounced effects on animal mobility. Considering that adequate neuronal and muscular activity is essential for proper animal locomotion, these results suggest that nanoplastic exposure may affect the neurophysiology in *C. elegans*.

3.2. Diverse effects of charged nanoplastics: physiology, locomotion, and bioaccumulation

We next investigated the effects of nanoplastics with different surface groups because weathered plastics could acquire surface charges and many studies have suggested differential effects of surface charges on tiny plastic particles [42–44,54,55]. We used three types of 25 nm

PS-NPs: pristine (PS), carboxyl (COOH)-modified PS (PS-COOH, which is expected to be dominantly negatively charged in the intestine of *C. elegans* at an average pH of 4.4), and primary amino (NH₂)-modified PS (PS-NH₂, which is expected to be dominantly positively charged in the weakly acidic intestinal lumen of *C. elegans*) [2,13]. Our findings showed that exposure to either PS or PS-COOH led to a decrease in body length (Fig. 2A), while PS-NH₂ exposure led to a decrease in thrashing frequency (Fig. 2B).

We further probed how these PS-NP, differing in surface charges, influenced tissue penetration dynamics. For this study, worms were exposed to red fluorescence-labeled PS (RF-PS) beads for short-term (1 day) and long-term (7 days) durations. During short-term exposure, all three nanoplastic bead variants were predominantly localized in the pharyngeal and intestinal lumens in all worms (Fig. 3A and B). Additionally, these particles were detected throughout the entire head region in some worms, with the most significant occurrence observed with PS-COOH beads (60% of worms) compared to pristine PS (22%) or PS-NH₂ (41%) (Fig. 3B). During long-term exposure, the beads not only remained in the pharyngeal lumen but also penetrated and spread across the entire head region in the majority of worms (Fig. 3A and B). Among the variants, PS-COOH showed the highest level of penetrance (Fig. 3B).

Following extended exposure, the nanoparticles exhibited a capacity to move beyond their initial entry points in the digestive tract and infiltrate deeper extraintestinal tissues. Such observations suggest potential bioaccumulation and long-term toxicological implications from nanoplastic exposure.

3.3. Diverse effects of charged nanoplastics: leaky gut and muscle mitochondrial fragmentation

We speculated whether such pronounced tissue penetration could induce a "leaky gut" condition (Fig. 4A). To investigate this, we conducted an intestinal barrier function assay using blue food dye. In the wild-type worm control, the blue food dye was detected only in the pharyngeal and intestinal lumens in the majority of worms (86.7%), while it was found throughout the worm body in 13.3% of worms, likely due to a leaky gut condition (Fig. 4A and B). We observed that exposure to PS-NH₂ resulted in the most significant leaky gut condition (Fig. 4B; 35.1%).

Additionally, we delved into whether the muscle cells, located further outside the gut tissue, were affected. Mitochondrial morphology is inherently dynamic; while typically tubular in form, it can swiftly transform to a fragmented state under cellular stress (Fig. 4C). We assessed the muscle mitochondrial morphology dynamics using the worm reporter strain, SD1347 (*ccIs4251[myo-3p::mitochondrial GFP (mitoGFP); myo-3p::GFP::LacZ::NLS]*) [30,37]. In the day 7 adult

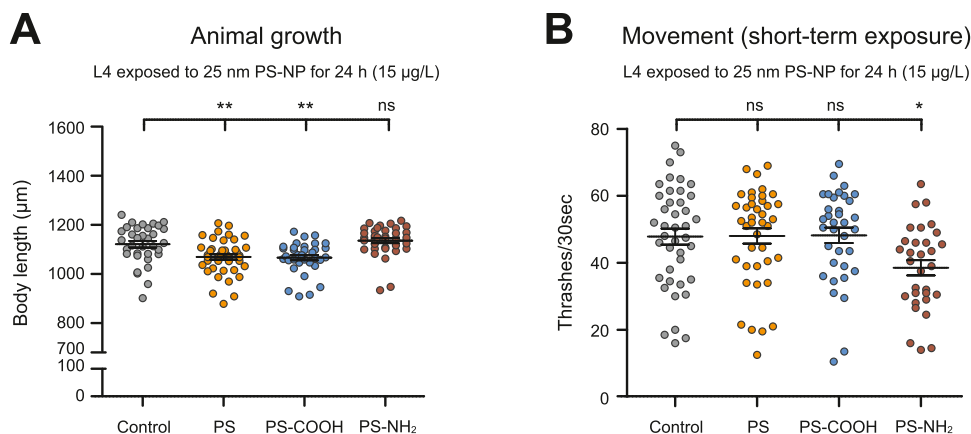


Fig. 2. Differential effects of PS-NP with varied surface charges on *C. elegans* physiology. (A) Body length of wild type *C. elegans* at the L4 stage after exposure to differently charged PS-NP. Statistics: one-way ANOVA. $N = 3$. $n = 34, 43, 35$, and 35 worms per group, respectively. (B) Thrashing frequency of wild type *C. elegans* at the L4 stage after exposure to PS-NP. Statistics: one-way ANOVA. $N = 3$. $n = 41, 40, 35$, and 31 worms per group, respectively.

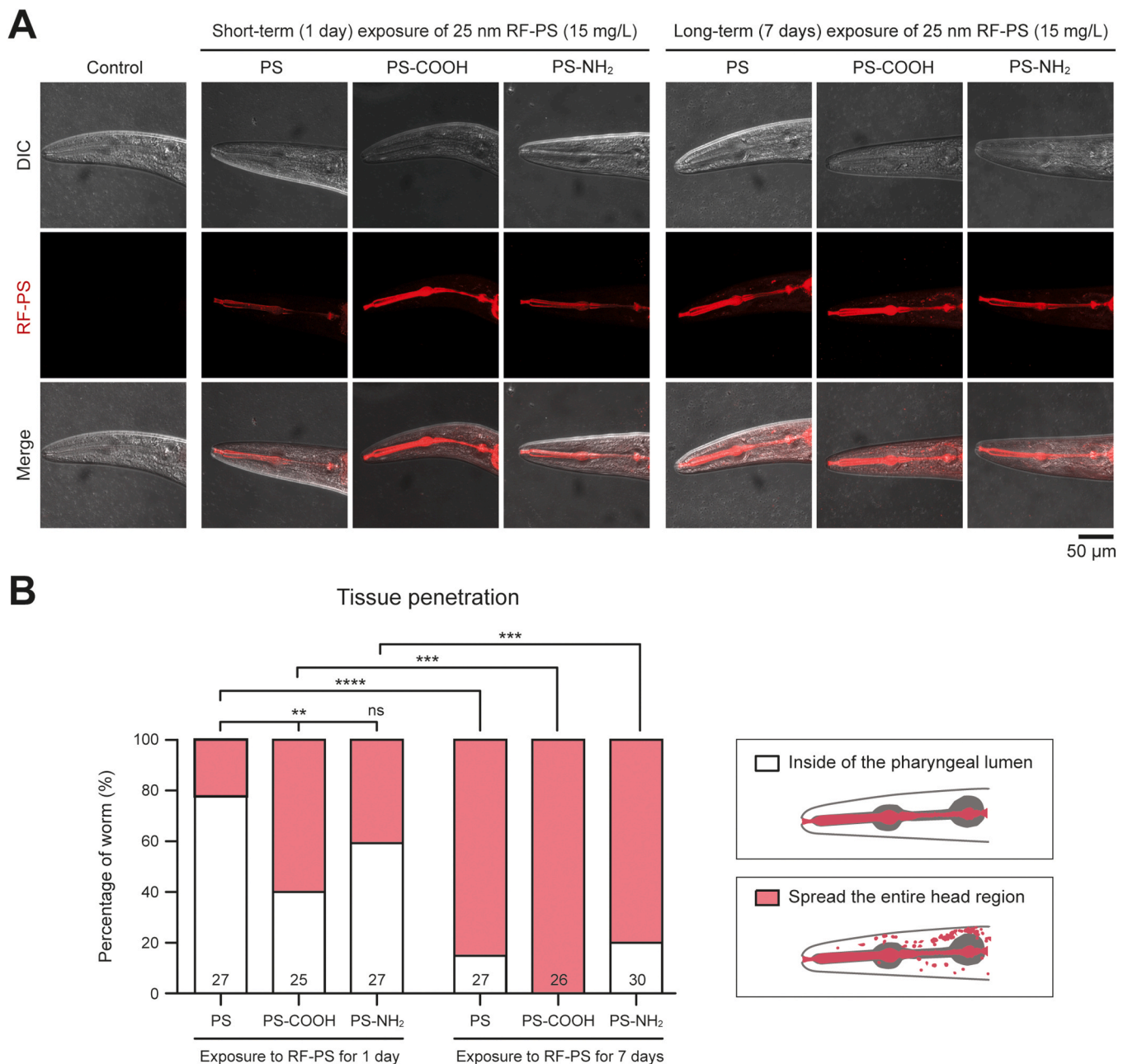


Fig. 3. Prolonged exposure to PS-NP results in tissue penetration in *C. elegans*. (A) Confocal images showing the spatial distribution of 25 nm red fluorescent PS beads (RF-PS) in *C. elegans* after 1-day and 7-day exposures. Images processed for maximum intensity using orthogonal projections. (B) Quantitative assessment of tissue penetration rates of 25 nm RF-PS beads. Data shown as percentages. Statistics: Fisher's exact test. $N = 3$. The total sample size is indicated in the bar.

wild-type worm controls, only 25% of worms showed fragmented mitochondria in muscle, while 57.7% of worms exhibited fragmentation when exposed to PS-NH₂ beads (Fig. 4D). It is consistent with the increased frequency of the leaky gut condition following PS-NH₂ exposure (Fig. 4B).

Given that mitochondrial fragmentation is frequently associated with elevated oxidative stress [60], we assessed the oxidative stress levels using the worm reporter strain, CL2166 (*dvIs19[gst-4p::GFP::NLS]*). This strain is designed to visually indicate oxidative stress levels by expressing GFP under the *gst-4* promoter, a known oxidative stress-responsive element [21]. Contrary to our expectations, there was no increase in oxidative stress following PS-NP exposure (Fig. 4E and F). This suggests that the physiological effects of nanoplastics may not solely rely on well-understood mechanisms like oxidative stress. There might be other avenues of nanoplastic-induced cellular stress and

subsequent physiological outcomes to consider.

In summary, we identified distinct responses to nanoplastics depending on their surface charge. PS-NH₂ beads, positively charged nanoplastics, induced leaky gut, compromising the integrity of the intestinal barrier (Fig. 4B). Furthermore, such nanoplastics prompted mitochondrial fragmentation in muscles (Fig. 4D), which correlates with the observed reduction in locomotion upon PS-NH₂ exposure shown in Fig. 2B. These findings highlight the intensified cellular stress and potential toxic effects posed by charged nanoparticles.

3.4. Associations between nanoplastic exposure and manifestations of PD symptoms in *C. elegans* PD models

Recent studies highlight that “leaky gut” or increased intestinal permeability might contribute to the onset and progression of PD,

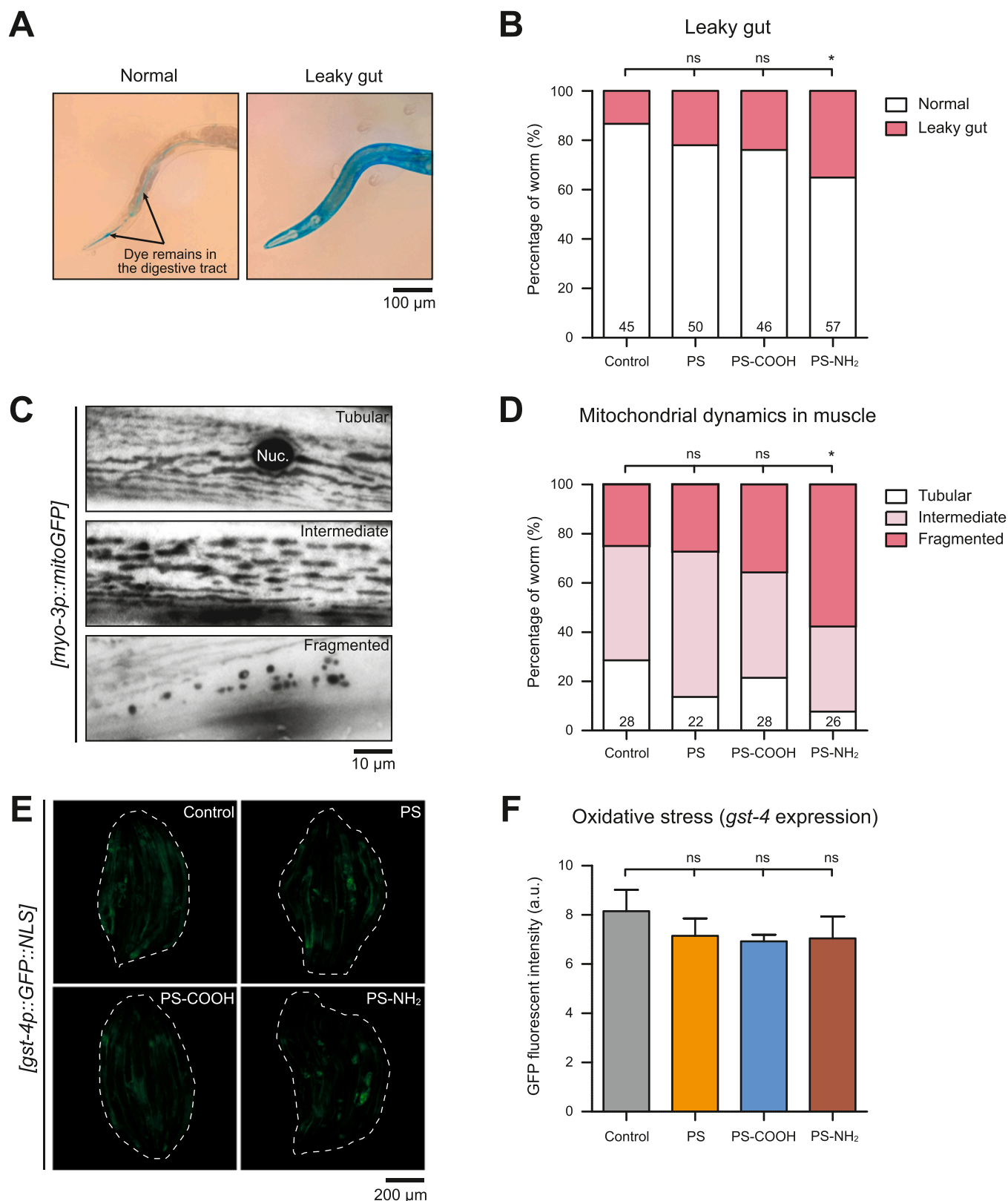


Fig. 4. Exposure to 25 nm PS-NH₂ beads induces leaky gut and mitochondrial fragmentation in *C. elegans*. (A) Intestinal barrier integrity assay in wild type *C. elegans*. In each experiment, embryos were exposed to PS-NP at a concentration of 15 μ g/L for 7 days. (B) Distribution percentages of worms displaying leaky gut condition. Data shown as percentages. Statistics: Fisher's exact test. $N = 4$. The total sample size is indicated in the bar. (C) Visualization of three distinct mitochondrial morphologies using the SD1347 strain. (D) Proportional representation of worms presenting each mitochondrial morphology. Data shown as percentages. Statistics: Fisher's exact test (Comparing 'Tubular' and 'Intermediate' vs. 'Fragmented'). $N = 3$. The total sample size is indicated in the bar. (E) Visualization of oxidative stress response using the CL2166 strain. (F) Quantitative representation of GFP fluorescence intensity, denoted in arbitrary units (a.u.). Statistics: one-way ANOVA. $N = 3$. $n = 40$ worms per group in each experiment.

potentially through mechanisms related to gut-brain axis, systemic inflammation, α -Synuclein aggregation [56]. In this context, we explored the effects of 25 nm PS bead exposure in two *C. elegans* models of PD. We first used UA44 (*bal11[dat-1p:: α -Synuclein; dat-1p::GFP]*) strain, a well-characterized genetic model of PD that overexpresses human α -Synuclein in dopaminergic neurons [47]. In this *C. elegans* PD model, dopaminergic neurons degenerate at a faster rate compared to

wild-type worms [58]. We sought to determine whether prolonged exposure to 25 nm PS-NP exacerbates neurotoxic effects. We examined all three types of nanoplastics with different surface charges and discovered that each of them exacerbated the degeneration of dopaminergic neurons, particularly when observing the CEP and ADE neurons in the head region (Fig. 5A and B). These dopaminergic neurons exhibited multiple signs of degeneration, such as neurite blebbing, a

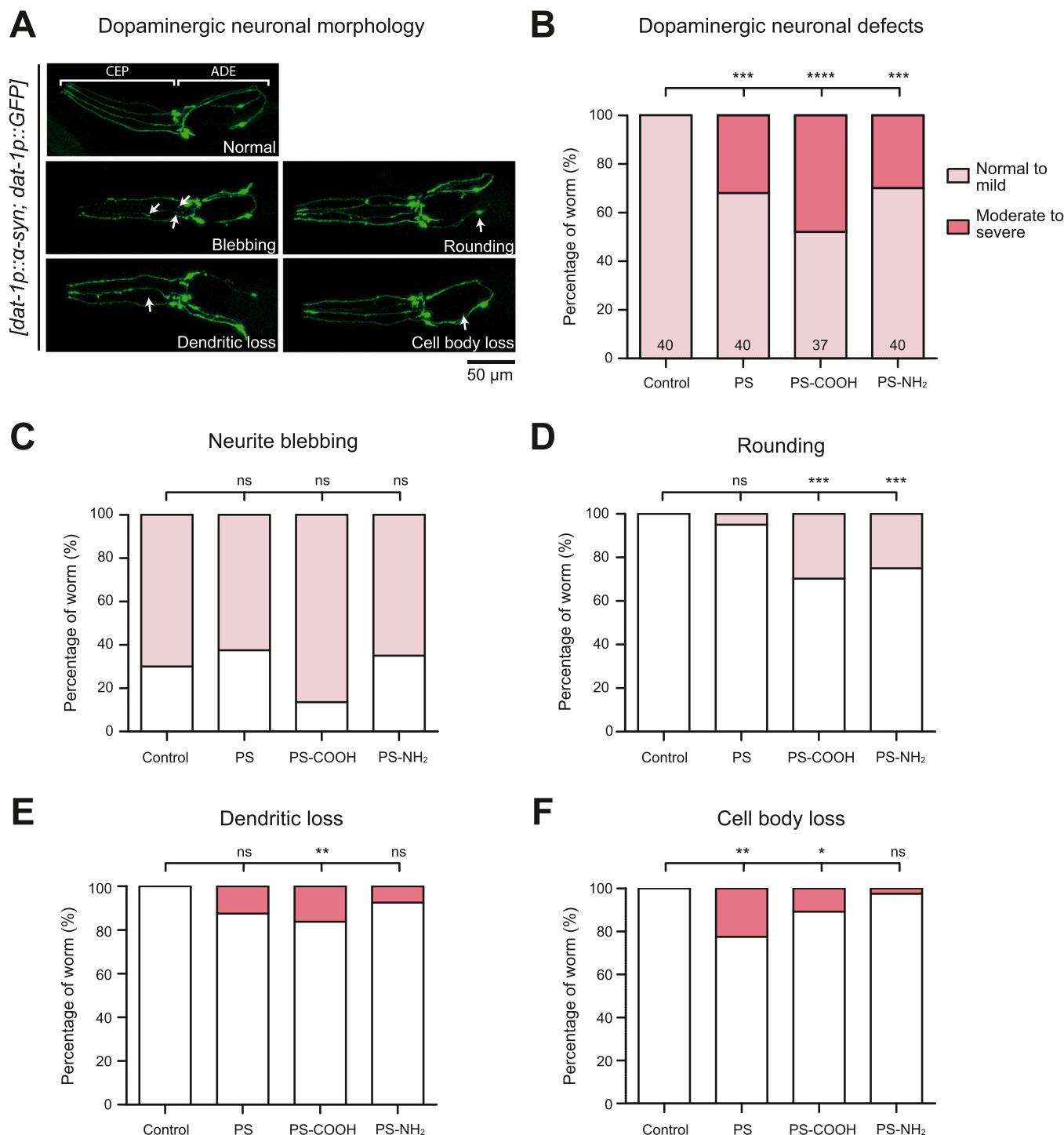


Fig. 5. Exposure to PS-NP induces dopaminergic neuronal degradation in the *C. elegans* PD model. (A) Visualization of the UA44 strain to illustrate various types of dopaminergic neuronal damage. Specific abnormalities are indicated by arrowheads. Embryos were exposed to PS-NP at a concentration of 15 μ g/L for 7 days. (B-F) Quantitative assessments detailing each type of neuronal damage resulting from exposure to PS-NP. (B) The phenotype varies in severity, ranging from mild, which characterized by blebbing or rounding, to moderate and severe, which displays a combination of four types of defects: neurite blebbing (C), cell body rounding (D), dendritic loss (E), and cell body loss (F). Data shown as percentages. Statistics: Fisher's exact test. $N = 3$. The total sample size is indicated in the bar.

rounded appearance of the ADE neuron cell body, dendritic loss, and the disappearance of the ADE cell body (Fig. 5C-F).

We then investigated whether the 25 nm PS-NP beads induce the aggregation of the α -Synuclein protein, a hallmark characteristic of PD [52]. To investigate this, we used another worm PD model strain, NL5901 (*pkIs2386[unc-54p:: α -Synuclein::YFP]*), which expresses the human α -Synuclein protein in its body wall muscle (Fig. 6A). Notably, upon exposure to PS or PS-COOH, there was a dramatic increase in the number of α -Synuclein aggregates (Fig. 6B; a 2.3-fold increase), even though the individual aggregate size remained relatively unchanged (Fig. 6C). Consequently, the total area of aggregates expanded upon exposure to PS or PS-COOH (Fig. 6D; a 2.8-fold and 2.74-fold increase, respectively, compared to control).

We finally asked whether these worms have any locomotive impairments, particularly in the body movement, one of the key symptoms of PD. We found that pristine PS showed the most dramatic reduction in thrashing frequency (Fig. 6E; 56.2% reduction in frequency).

Collectively, these findings indicate a potential association between nanoplastic exposure and neurodegenerative alterations seen in PD, such as dopaminergic neuronal degeneration, the accumulation of α -Synuclein aggregates, and locomotor dysfunction [38,52]. Our study underscores the varied neurodegenerative impacts influenced by the distinct surface charges of nanoplastics. Specifically, exposure to pristine PS or negatively charged PS-COOH leads to an increase in α -Synuclein aggregates and pristine PS leads to a reduction in locomotion in *C. elegans* models of PD.

3.5. Associations between nanoplastic exposure and manifestations of PD symptoms in human cell line

To study the effects of nanoplastic particles in human cells, we utilized the A53T α -syn-EGFP SH-SY5Y cell line, derived from human neuroblastoma. To monitor the penetration of pristine PS into the cells, we first exposed the cells to RF-PS beads. Our observations showed that these nanoparticles penetrated the cells, with a predominant location in the cytosol (Fig. 7A). To establish an in vitro PD model, we introduced α -Synuclein pre-formed fibrils (PFF) to initiate the formation of α -Synuclein aggregates in A53T α -syn-EGFP SH-SY5Y cells [22]. The use of A53T α -syn, a variant associated with familial PD, accelerated the aggregation process [10], and the incorporation of the EGFP tag facilitated efficient analysis [22]. In this PD model, we examined the effects of 25 nm pristine PS exposure on α -Synuclein aggregation and found that pristine PS further increased the accumulation of α -Synuclein in the A53T α -syn-EGFP SH-SY5Y cells (Fig. 7B and C). Specifically, pristine PS resulted in a 50.7% increase in the number of α -Synuclein aggregates in each cell compared to the control with PFF (Fig. 7C). The correlation we found between nanoplastic exposure and PD development in our models may suggest potential implications for human pathology.

4. Discussion

The introduction of micro- and nano-plastics into the environment [20], primarily due to human activities, presents a growing concern for the health of both the environment and living organisms [12]. Our study uses *C. elegans* as a model system to investigate the potential physiological and neurological consequences of nanoplastic exposure in organisms, especially those derived from polystyrene nanoparticles. The chosen model organism, renowned for its conserved metabolic pathways and physiological characteristics [11], allows us to gain valuable insights into potential mechanisms of nanoplastic toxicity.

One of the surprising results is the increased physiological response at lower concentrations such as 10 μ g/L or 15 μ g/L (Fig. 1A), implying that even trace amounts of nanoplastics can be harmful. Given that the concentration of 50 nm plastic particles in the environment is reported to be estimated between 1 pg/L and 15 μ g/L [1,32], it is possible that the current environmental exposure may already be harmful to many

organisms. Then, how does the relatively low concentration of PS-NP lead to more harmful effects? The exact mechanism underlying this observed phenomenon is yet to be fully elucidated. Some studies showed nanoplastics form aggregates with *E. coli*, a primary food source for *C. elegans* and these aggregates tend to grow larger with increasing nanoplastic concentrations [19]. Considering the nematodes' preference for smaller prey [25], they might avoid larger aggregates present at higher nanoplastic concentrations. This could lead to higher ingestion of nanoplastics at lower concentrations, resulting in lower uptake of *E. coli*. Such a dietary shift may cause nutritional imbalances, adversely affecting nematode growth. Thus, in this case, the stronger effects of low concentration could be specific to small-sized animals like *C. elegans*.

The surface charge of the nanoplastic particles highlights the complexity of their interactions with biological systems. Notably, the effects of nanoplastics can differ depending on their specific chemical attributes. We find that both the pristine PS and the negatively charged PS-COOH hinder worm growth (Fig. 2), while the positively charged PS-NH₂ nanoparticles appears more likely to damage the intestinal barrier, which may lead to mitochondrial fragmentation in the muscles (Fig. 4), potentially contributing to the movement impairments (Fig. 2). The mechanisms behind the variable effects of nanoplastics based on their surface charges remain unclear. One speculative idea is that a relatively high percentage of NH₂ groups become ionized, forming cations within the weakly acidic *C. elegans* gut environment (note that the mammalian small intestine is typically alkaline), which may facilitate the excretion of such cationic PS with waste matter. Conversely, pristine PS and PS-COOH are more likely to remain in the gut and interfere with nutrient absorption, ultimately inhibiting worm growth (Fig. 2). However, cationic PS can be more readily internalized by cells due to electrostatic interactions with the negatively charged cell membranes, as we observed more damage to both intestinal barrier and muscle cells (Fig. 4). In the worm models of PD, PS-NP particles with different surface groups exacerbate defects in dopaminergic neurons, while showing subtle differences in morphological defects (Fig. 5). On the other hand, pristine PS and PS-COOH induce a more significant accumulation of α -Synuclein aggregates (Fig. 6). Among them, pristine PS causes the most significant movement impairments (Fig. 6). The underlying mechanism explaining these differences based on the surface groups of PS-NP remains unclear. One speculation is that the positively charged N-terminal domain of α -Synuclein [14] may have a stronger preference for interacting with PS-COOH or pristine PS, leading to the easier formation of aggregates, rather than with PS-NH₂. There is a growing body of research on bioaccumulation and biological effects driven by the physicochemical properties of tiny plastic particles [34,42–44,54,55]. This emphasizes the importance of understanding the physicochemical properties of nanoplastics when assessing their potential hazards.

Our observations linking nanoplastic exposure to symptoms reminiscent of PD are particularly alarming. The neuropathology of PD is notably characterized by the presence of pathological inclusions of α -Synuclein, referred to as Lewy bodies [17], which are associated with degeneration of dopaminergic neurons [6]. While previous studies had suggested a connection between nanoplastics and altered gene expression related to PD in mouse neurons [35], our study provides a more comprehensive understanding of this relationship. We demonstrate that nanoplastic exposure can exacerbate the pathological manifestations of PD, such as dopaminergic neuronal degeneration and the accumulation of α -Synuclein aggregates in *C. elegans* models of PD (Figs. 5 and 6). The significance of our study is further emphasized by the results from the human cell model of PD, a well-established cell model for evaluating α -Synuclein aggregation (Fig. 7). The observed α -Synuclein aggregation upon nanoplastic exposure in human cells makes the extrapolation of our findings from *C. elegans* to potential human effects even more relevant. Given that increased α -Synuclein aggregates are a hallmark of PD [52], post-nanoplastic exposure is particularly concerning. These aggregates play a central role in PD pathophysiology, and any factor exacerbating their formation or accumulation needs to be critically

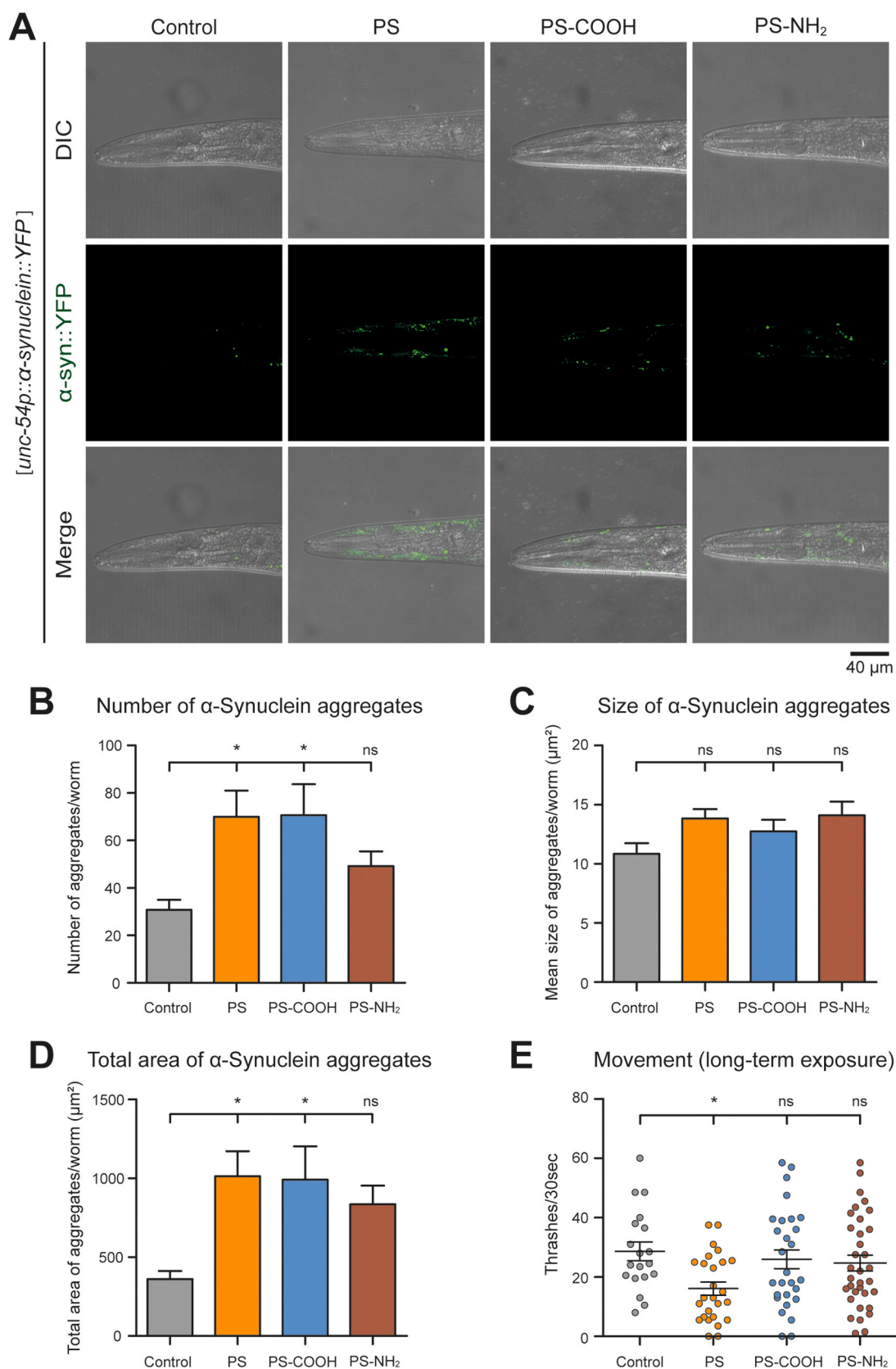


Fig. 6. Exposure to PS-NP increases α -Synuclein aggregates in *C. elegans*. (A) Confocal images of the NL5901 strain treated with three different types of 25 nm PS beads. Imaging optimized using orthogonal projections (standard deviation method). Region of interest (ROI) spans from the nose-tip to the posterior of the terminal pharyngeal bulb. Embryos were exposed to PS-NP at a concentration of 15 μ g/L for 7 days. (B-D) Quantitative assessments of α -Synuclein aggregate characteristics. Statistics: one-way ANOVA. $N = 4$. $n = 50$ worms per group. (E) Thrashing frequency of the NL5901 strain after exposure to PS-NP at a concentration of 15 μ g/L for 7 days. Statistics: one-way ANOVA. $N = 3$, $n = 19$, 26, 28, and 34 worms per group.

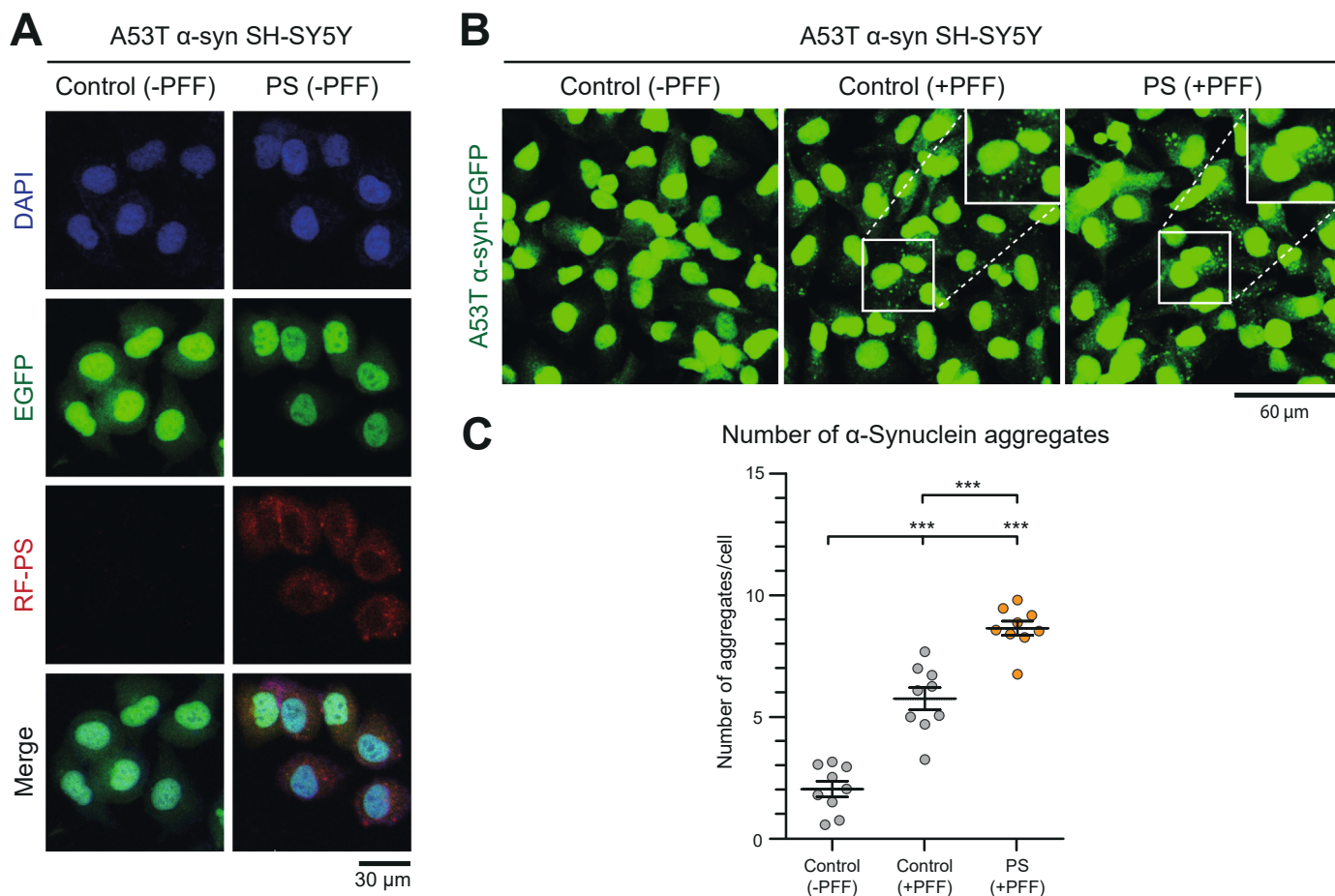


Fig. 7. Exposure to PS-NP leads to their penetration into human cells and results in an increase in α -Synuclein aggregation in the human cell model of PD. (A) Confocal images showing the spatial distribution of 25 nm RF-PS beads in SH-SY5Y cells after 1-day exposure. (B) Confocal images of A53T α -syn-EGFP SH-SY5Y cells. These cells were treated with 250 nM PFF, either with or without 15 mg/L of PS-NP for one day. (C) Quantification of the number of α -Synuclein aggregates per cell. Statistics: one-way ANOVA. $N = 3$. $n = 3$ processed images were analyzed, with each image containing 37–80 cells per group.

evaluated. In this context, we provide critical evidence that nanoplastics act as exacerbating factors in the aggregation of α -Synuclein, thereby contributing to the PD pathology in both nematodes and human brain cells.

What are the ways in which exposure to nanoplastics contributes to the development of PD? The gut has recently been highlighted as a significant starting point for α -Synuclein aggregation and the formation of Lewy bodies, the pathological inclusions of α -Synuclein [17,36]. These aggregates can potentially spread to the central nervous system via the vagus nerve, thereby contributing to the development of PD [7]. Recent findings indicate that when nanoplastic particles are consumed, they can trigger an immune response in the gut, impacting brain immunity and leading to cognitive deficits in mice given nanoplastics in their diet [62]. Furthermore, there is mounting evidence suggesting that nano- and micro-plastics could compromise the human intestinal barrier's integrity, as they have been detected in both human feces and blood [33,48,65]. Consistent with these findings, our study also shows that nanoplastic particles induce leaky gut conditions, infiltrate extra-intestinal tissues (Figs. 3 and 4), and further affect neuronal morphology, levels of α -Synuclein aggregates, and locomotion (Figs. 5 and 6). Thus, one possibility for how nanoplastics affect the nervous system is that the penetrating nanoparticles reach the neurons directly through impaired intestinal integrity, while another possibility is that the intestinal immune response affects the neurons via the gut-brain axis. These possibilities are not mutually exclusive, and other models are, of course, possible. Regardless of the actual mechanisms, this raises the concern that nanoparticles ingested through the gut or inhaled

through the lungs in humans may also impact neuronal health and integrity.

In summary, tiny plastic particles could increase the risk of neurodegenerative diseases such as PD. Our findings offer significant insights into the potential health hazards of nanoplastics, it also underscores the need for further investigations. While *C. elegans* offers a robust model system, extrapolating our findings to more complex organisms, including humans, requires caution. Additionally, while we observed exacerbation of PD-like symptoms upon nanoplastic exposure, the molecular mechanisms underlying these observations remain elusive. Future studies must aim at elucidating these molecular pathways.

5. Conclusion

Our study is among the pioneering efforts highlighting the potential neurotoxic effects of nanoplastics. We provide compelling evidence for the deleterious effects of 25 nm PS-NP on the neurophysiology of *C. elegans* and demonstrate their potential to exacerbate Parkinson's-like symptoms. The variable impacts based on nanoplastic concentration and surface charge emphasize the multifaceted nature of nanoplastic toxicity. As the prevalence of these nanoplastics in the environment continues to grow, understanding their effects on health becomes paramount. While our study sets the stage for understanding these interactions, it also underscores the pressing need for more comprehensive studies and increased scrutiny of plastic pollutants.

CRedit authorship contribution statement

Ayoung Jeong: Conceptualization, Methodology, Data curation, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Soo Jung Park:** Investigation, Formal analysis, Visualization, Writing – original draft. **Eun Jeong Lee:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing, Supervision. **Kyung Won Kim:** Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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